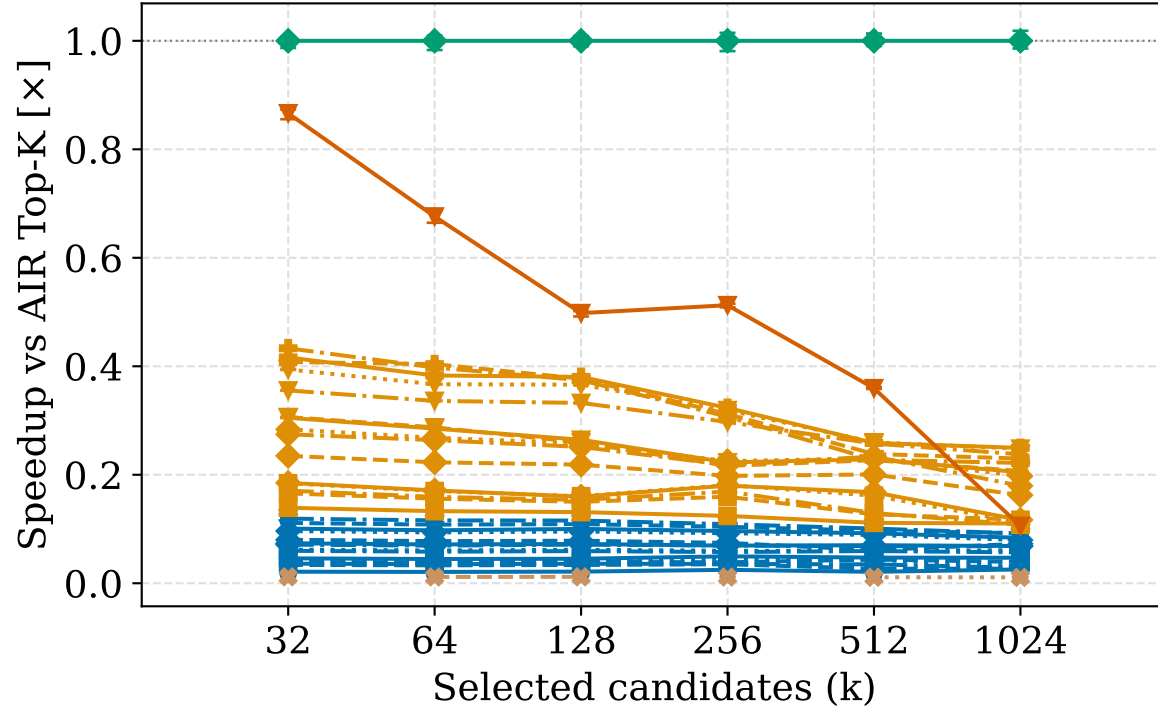
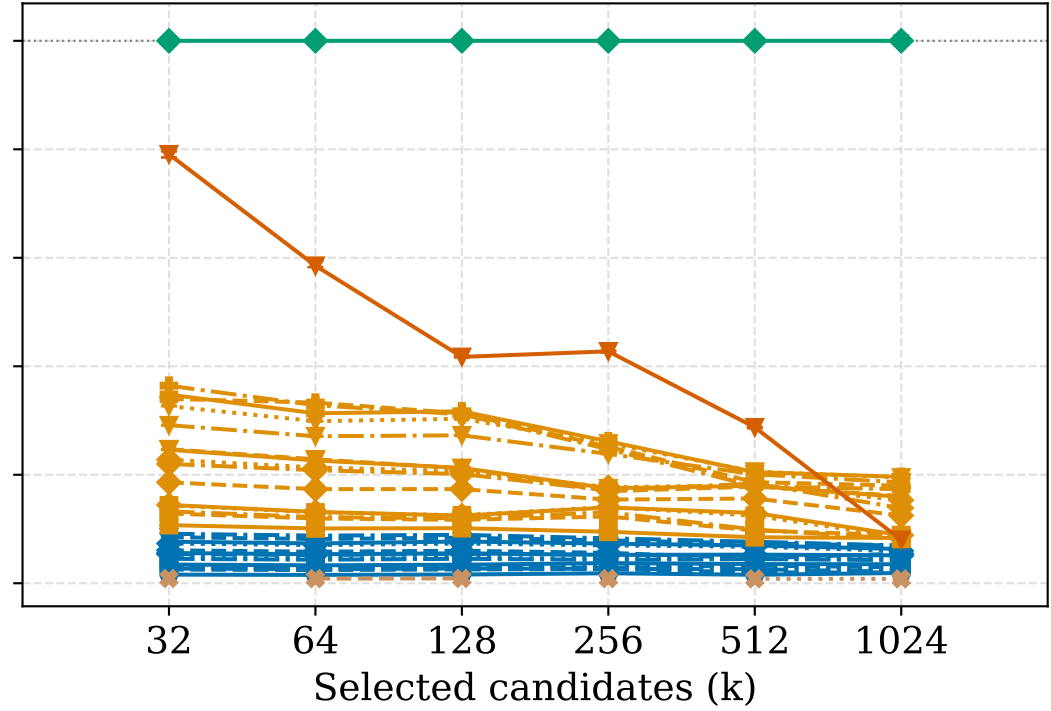


gradient-compression: 2,359,296 candidates; batch=1; input e77ffd0b75; split degree=8  
 gradient-compression-middle — AIR median / backend median; >1 is faster  
 Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



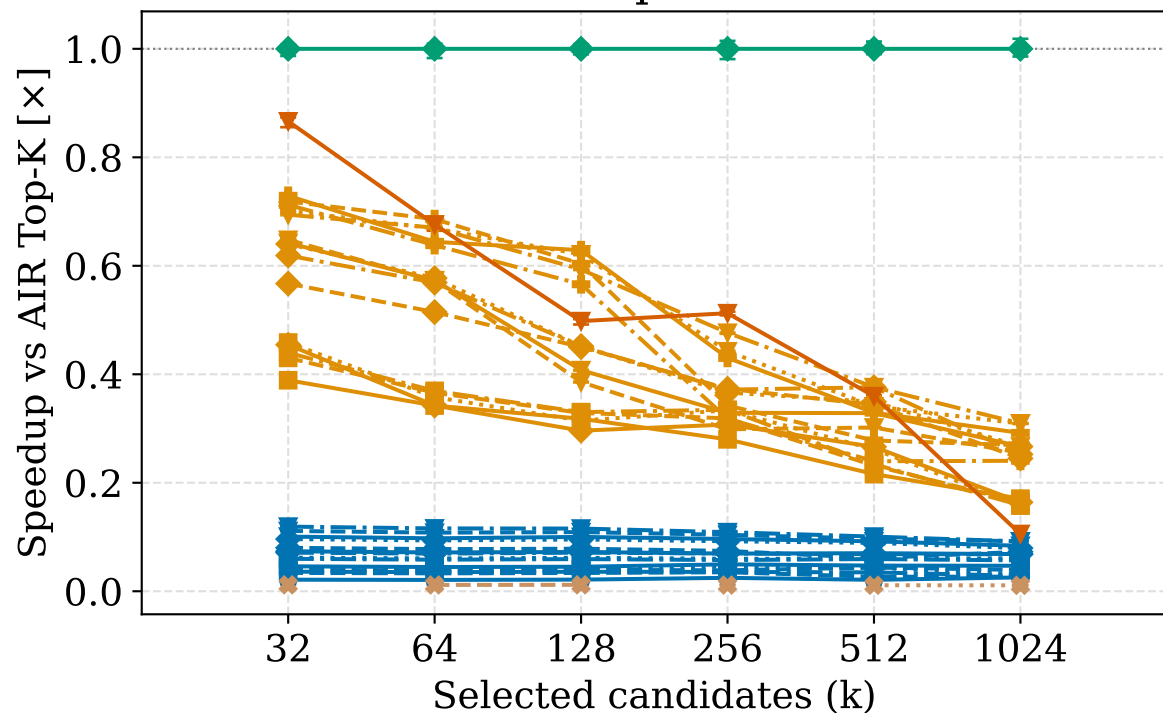
- |                                             |                                              |
|---------------------------------------------|----------------------------------------------|
| bits (degree=1, block=128, items=4)         | bits (split) (degree=8, block=128, items=13) |
| bits (degree=1, block=128, items=7)         | bits (split) (degree=8, block=128, items=16) |
| bits (degree=1, block=128, items=8)         | bits (split) (degree=8, block=256, items=4)  |
| bits (degree=1, block=128, items=13)        | bits (split) (degree=8, block=256, items=7)  |
| bits (degree=1, block=128, items=16)        | bits (split) (degree=8, block=256, items=8)  |
| bits (degree=1, block=256, items=4)         | bits (split) (degree=8, block=256, items=13) |
| bits (degree=1, block=256, items=7)         | bits (split) (degree=8, block=256, items=16) |
| bits (degree=1, block=256, items=8)         | bits (split) (degree=8, block=512, items=4)  |
| bits (degree=1, block=256, items=13)        | bits (split) (degree=8, block=512, items=7)  |
| bits (degree=1, block=256, items=16)        | bits (split) (degree=8, block=512, items=8)  |
| bits (degree=1, block=512, items=4)         | bits (split) (degree=8, block=512, items=13) |
| bits (degree=1, block=512, items=7)         | bits (split) (degree=8, block=512, items=16) |
| bits (degree=1, block=512, items=8)         | AIR Top-K (degree=1, block=512, items=0)     |
| bits (degree=1, block=512, items=13)        | GridSelect (degree=1, block=0, items=0)      |
| bits (degree=1, block=512, items=16)        | BlockSelect (degree=1, block=128, items=2)   |
| bits (split) (degree=8, block=128, items=4) | BlockSelect (degree=1, block=128, items=3)   |
| bits (split) (degree=8, block=128, items=7) | BlockSelect (degree=1, block=128, items=4)   |
| bits (split) (degree=8, block=128, items=8) | BlockSelect (degree=1, block=128, items=8)   |

gradient-compression: 2,359,296 candidates; batch=1; input e77ffd0b75; split degree=32

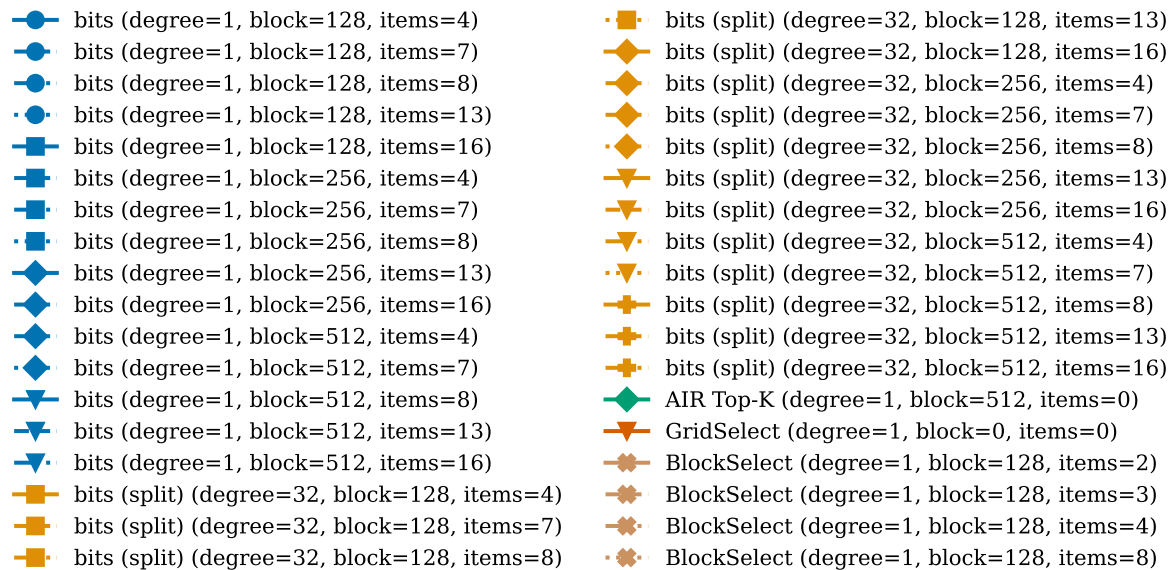
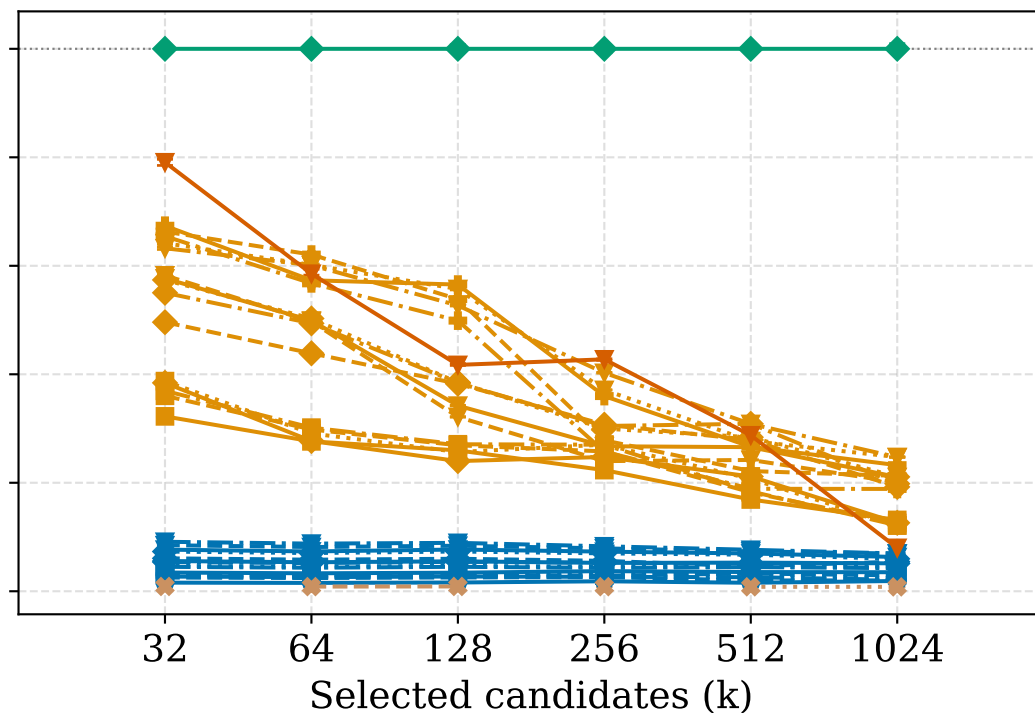
gradient-compression-middle — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection

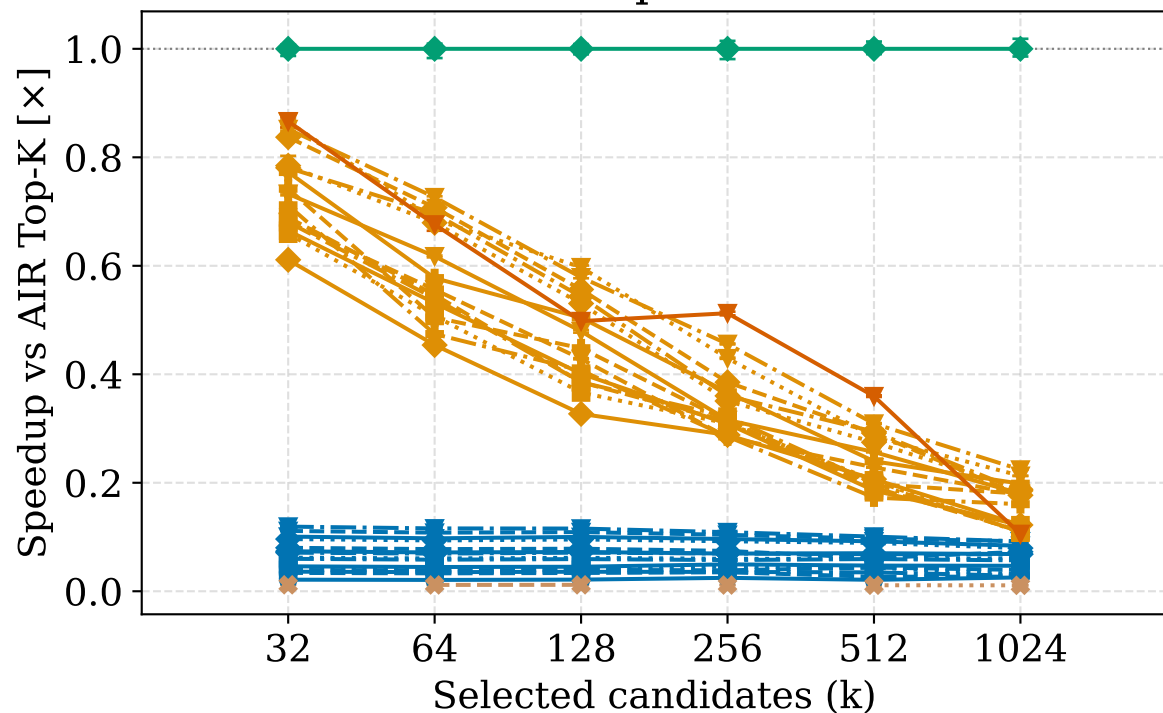


gradient-compression: 2,359,296 candidates; batch=1; input e77ffd0b75; split degree=128

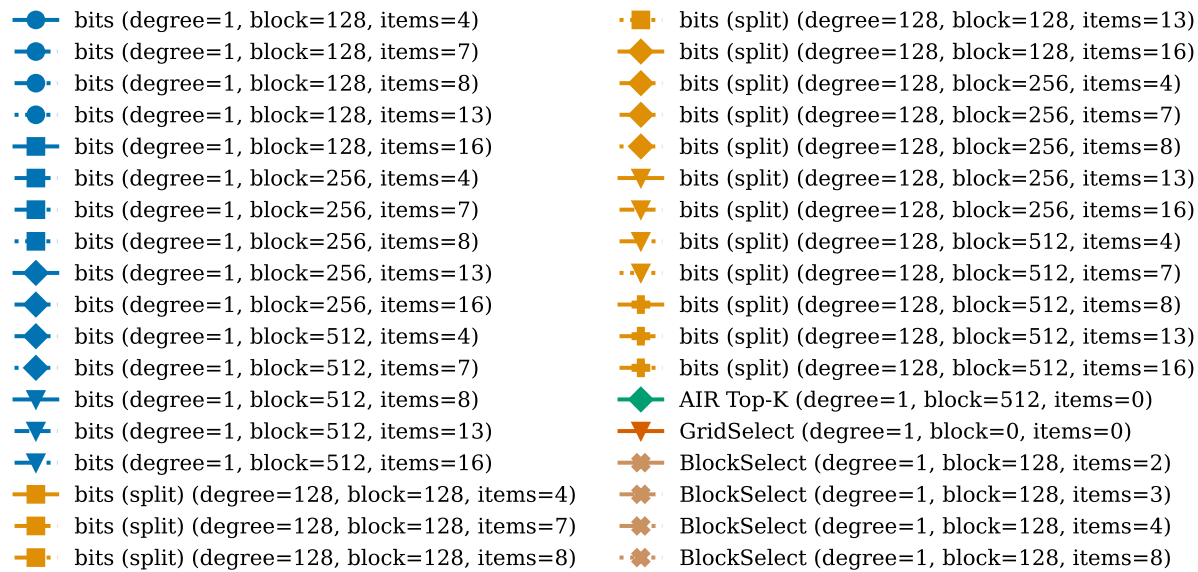
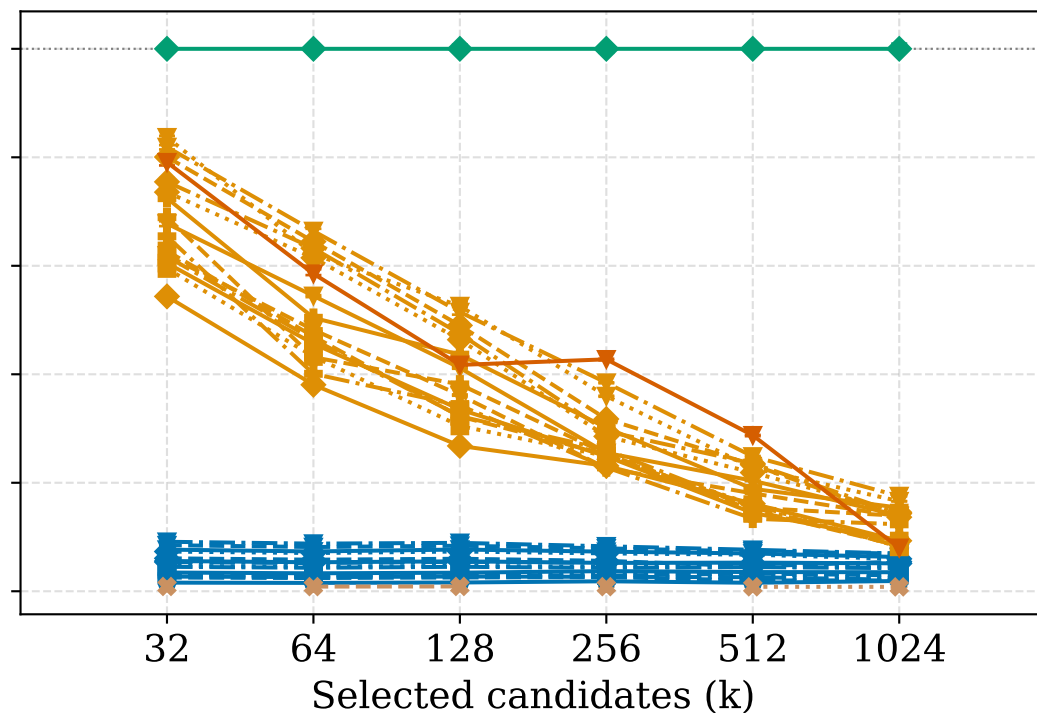
gradient-compression-middle — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

### Full operator



### Isolated selection

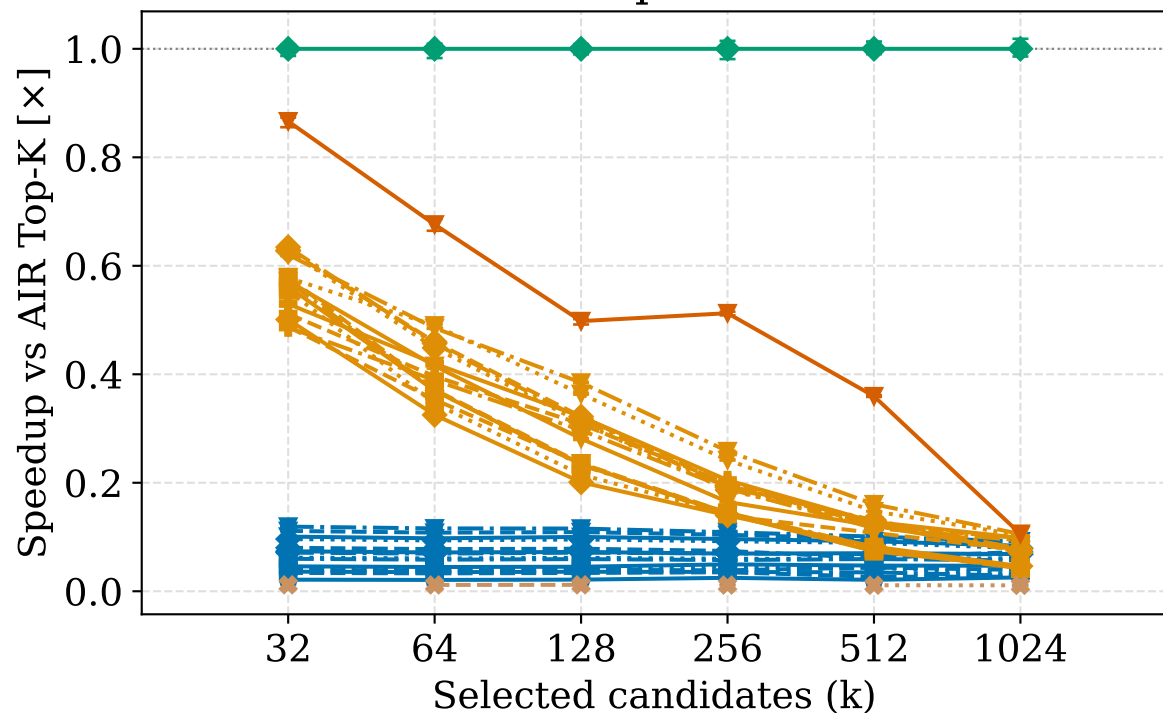


gradient-compression: 2,359,296 candidates; batch=1; input e77fd0b75; split degree=512

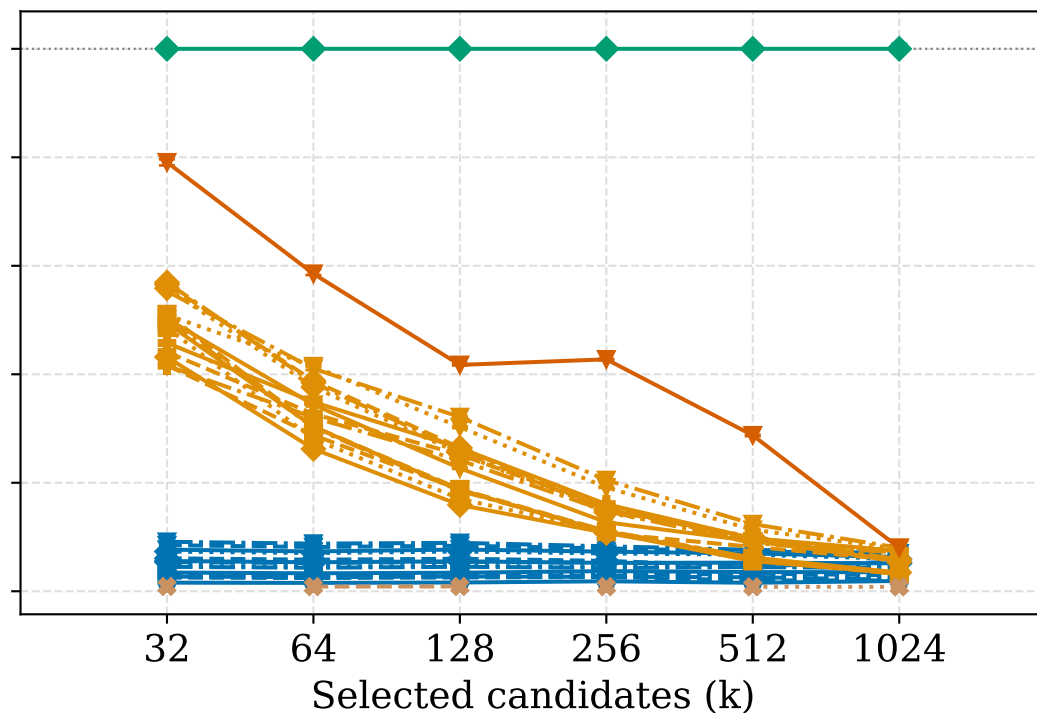
gradient-compression-middle — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



- bits (degree=1, block=128, items=4)
- bits (degree=1, block=128, items=7)
- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=512, block=128, items=4)
- bits (split) (degree=512, block=128, items=7)
- bits (split) (degree=512, block=128, items=8)
- bits (split) (degree=512, block=256, items=4)
- bits (split) (degree=512, block=256, items=7)
- bits (split) (degree=512, block=256, items=8)
- bits (split) (degree=512, block=256, items=13)
- bits (split) (degree=512, block=256, items=16)
- bits (split) (degree=512, block=512, items=4)
- bits (split) (degree=512, block=512, items=7)
- bits (split) (degree=512, block=512, items=8)
- bits (split) (degree=512, block=512, items=13)
- bits (split) (degree=512, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)